Math **163** Name (PLEASE PRINT) Srinarayan Srikanth

Exam **1B** **2026** Fall/Spring/Summer I /**Summer II** Score 0

Please fill in the following table putting in expected points corresponding to the parts of problems that you have completed and submitted and an expected total. Leave blank those parts that you have not completed or that do not exist. You will lose double the points that a part is worth if you submit incomplete, unproved, or unchecked answers, or if you do not follow the directions of the problem. Ideally you should know exactly what you have done correctly and how many points you should receive for parts and in total. I will fill in the received grade after I correct the exam. Not all questions have the same number of parts so that some columns may be redundant for particular problems.

										Expected	Received
	Part	Part	Part	Part	Part	Part	Part	Part	Part	Grade	Grade
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)		
Number											
1abc											
2a											
2b											
2c											
3a											
3b											
3c											
								Total			

Instructions

- All work must be done using a **MECHANICAL PENCIL following instructions given**.
- If you have any questions at all, please **ask me**. This includes the following possibilities:
    - something is **illegible,** or **ambiguous**, or **unclear** to you.
- **You may use your homework, notes, books, and calculators unless otherwise stated during the exam.**
- **You will not get any partial credit**. You have to show **absolutely all your work** on all questions, **exactly as in the notes, with no changes, variations, or omissions in form, unless otherwise noted whether orally or in writing**. If your answer is correct, but your procedure is incorrect or is not allowed by the instructions, you will get no credit all. All work must be displayed using **correct syntax**, and all **explanations**, written in **complete sentences**, must be complete.
- **You must write neatly, clearly and legibly. You must draw every straight-line segment other than those in the list: (**$=, \neq, \equiv, <, \leq, >, \geq, +, -, \times, \div, \ \neg, \vee, \wedge, \parallel, \perp, \setminus, \rightarrow, \leftrightarrow, \mapsto$**) with your ruler. At least 1 point will be taken off for each instance of violation. Signs of equality or equivalence must line up, one below the other on the left of expressions. Every symbol has a specific meaning and may be used only in this sense and in no other. Syntax defined in the notes must be followed exactly to the letter.**
- Remember that 'does not make sense' is a possible answer.
- I can grade you only on what you **actually** put on paper; I cannot read minds. You cannot say later that what you had written down is not what you had meant. **You need to learn to write exactly what you mean and mean exactly what you write**.
- **The answers for a problem should be noted on the page on which the problem appears in the areas indicated. All work for the problem should be shown on sheets that follow the page on which the problem appears numbered as follows. Work for a problem that occurs on page n should be on blank (not ruled) sheets after page n, numbered, n0, n1, n2 etc. on the lower right-hand corner as in this document. For example, work for a problem on page 502 should be on pages numbered 5020, 5021, 5022 etc. as needed. Work that corresponds to different parts must be separated from each other by striaght lines drawn with a ruler parallel to the edges of the paper and labelled correctly as in the Pages from Student Work as found under Files on CANVAS.**
- **You should submit a SINGLE document.**
- **You may NOT include pages that you have NOT attempted**.
- **As shown in the class means as shown in the notes.**

Instructions:

Use these instructions for the remainder of the exam.

Your J-number $JN :=$ _____ $:= (J_k | k \in 1..8)$

Hence

$J_1 =$ $J_2 =$ $J_3 =$ $J_4 =$

$J_5 =$ $J_6 =$ $J_7 =$ $J_8 =$

R For each of sets defined below **write the elements in increasing order**

D Define $\mathcal{U} := 0..9 = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

D Define $J := \{J_k \in 0..9 | k \in 1..8\} = \{$ $\}$

D Define $\forall n \in \mathbb{N}$ $Prm(n) :\Leftrightarrow n \text{ is a prime number}$

D Define $E := \left\{ J_k \in 0..9 \middle| Prm(k) \right\} = \{$ $\}$

k		**1**		**2**		**3**		**4**		**5**		**6**		**7**		**8**	
J_k																	
$Prm(k)$		***Y***		***Y***		***Y***		***Y***		***Y***		***Y***		***Y***		***Y***	
		N		***N***		***N***		***N***		***N***		***N***		***N***		***N***	

D $s := mnm_{k \in 1..8}\{J_k \in 0..9 | J_k \neq 0\} =$

D $b := mxm_{k \in 1..8}\{J_k \in 0..9 | k \in 1..8\} + 1 =$

D Define

$$A := \{0, s, b\} = \{\quad\}$$

D Define

$$B := \left\{ J_k \in 0..9 \middle| \cancel{Prm}\left(k \right) \right\} = \{\quad\}$$

k		**1**		**2**		**3**		**4**		**5**		**6**		**7**		**8**	
J_k																	
$\cancel{Prm}\left(k \right)$		***Y***		***Y***		***Y***		***Y***		***Y***		***Y***		***Y***		***Y***	
		N		***N***		***N***		***N***		***N***		***N***		***N***		***N***	

D A partial order on a set is said to be total if and only if every pair of elements is comparable under the ordering. This word appears in 2c and part of the test is to determine if you can formulate this definition precisely using proper symbols and quantification and then use this definition in 2c.

D Define Y:= Yes N:=No

D Define Pf:= Proof W:=Witness

R **Note that in a set an element may occur exactly once**. Therefore, there may not be any repeated elements in either *A* or *B* in the above. **The sets *A* and *B* will occur in several problems on this test**. **So, if you do not have the correct sets, all your answers will be wrong.**

R If answers to different instances of some question on the exam contradict each other you will get a score of zero for every instance.

R **You will lose points for making any extraneous mark on the exam.**

R You may only submit complete answers to questions marked explicitly on the exam. You will lose points for unfinished work. **Do not write anything down on any page if you cannot completely solve the problem or some part thereof that is indexed by a roman numeral.**

R Separate pieces of work for any part of any problem recorded on a blank page intended for the purpose must be indexed by the index of the problem and also by numbers starting at 0 that record the order in which the pieces were completed; such parts must be separated from each other by straight lines parallel to the edges of the paper drawn with a ruler. You will lose points otherwise.

R Any sloppiness, untidiness, and any departure from proper format (as indicated in class) will lead to a score of 0.

R **In directed graphs, put the vertices in circles.**

Your J-number $JN \coloneqq$ _____

We define $\forall k \in 1..8$,

D $J_k \coloneqq$ the k[th] entry in your J-number counting from the left.

D $r_k \coloneqq$ the remainder obtained upon dividing J_k by 5.

Therefore,

$J_1 \coloneqq$

$J_2 \coloneqq$

$J_3 \coloneqq$

$J_4 \coloneqq$

$J_5 \coloneqq$

$J_6 \coloneqq$

$J_7 \coloneqq$

$J_8 \coloneqq$

Show all long divisions on the following page

$r_1 \coloneqq$

$r_2 \coloneqq$

$r_3 \coloneqq$

$r_4 \coloneqq$

$r_5 \coloneqq$

$r_6 \coloneqq$

$r_7 \coloneqq$

$r_8 \coloneqq$

D Define a relation $R \subseteq \{0,1,2,3,4\} \times \{0,1,2,3,4\} =: \mathcal{U}$ as follows:

Compute *R* on the following side, and write the answer below:

$$R$$

$$:= \left\{(0,0)\right\} \cup \left(\bigcup_{k \in 1..4} \left\{(r_{1+k}, r_{2+k})\right\} \right)$$

$$=$$

R **In order to finish this exam you need to learn by yourself how to multiply matrices**. Schaum's Outlines and the Internet contain information on matrices and may be consulted.

You will need the following definitions.

D Given the matrix for a relation $M(R)$ we define:

$$\big(M(R)\big)(0,1)$$

$$:= \quad Matrix\ obtained\ by\ replacing\ all\ non-zero\ entries\ in\ M(R)\ by\ 1$$

R **Note that in a set an element may occur exactly once**. Therefore, there may not be any repeated elements in either A or B in the above. **The sets *A* and *B* will occur in several problems on this test**. **So, if you do not have the correct sets, all your answers will be wrong.**

R **In arrow-diagrams, put the numbers in rectangles rather than ovals and draw the rectangles using rulers.**

R **Any sloppiness, untidiness, and any departure from proper format (as indicated in class) will lead to a score of 0**.

R Do not write anything down on any page if you cannot completely solve the problem or some part thereof that is indexed by a roman numeral.

17 (1abc) Carry out the following instructions, **doing all computations on pages that follow, properly numbered,** and **recording only the answers on this side**.

1 (i) **Draw directed graph for the relation *R*** earlier **in the space below** with 0, 1, 2, 3, 4 as nodes **arranged in the shape of a regular pentagon with a horizontal base**, with the left vertex of the horizontal side labelled 0, and labelling all the vertices in order traversing them in the counter-clockwise sense.

1 (ii) Find the 5× 5 matrix $M(R)$ for the relation R.

$$M(R) =$$

1 (iii) Find the 5× 5 matrix $M(R^{op})$ for the relation R^{op}.

$$M(R^{op}) \quad =$$

17 (1abc) Carry out the following instructions

5 (iv) Compute:

$$R \circ R^{op}, \qquad R^{op} \circ R,$$

$$M(R \circ R^{op}), \qquad M(R^{op} \circ R),$$

$$\big(M(R)\big)\big(M(R^{op})\big), \qquad \big(M(R^{op})\big)\big(M(R)\big)$$

Prove on **this side** that:

(1) $$\Big(M(R \circ R^{op})\Big)(0,1) = \Big(\Big(M(R^{op})\Big)\Big(M(R)\Big)\Big)(0,1)$$

(2) $$\Big(M(R^{op} \circ R)\Big)(0,1) = \Big(\Big(M(R)\Big)\Big(M(R^{op})\Big)\Big)(0,1)$$

17 (1abc) Carry out the following instructions

1 (v) Draw:

$$DG(R), \qquad DG(RflxClsr(R)) \qquad \text{and}$$

find: $$M\left(RflxClsr(R)\right), \qquad M\left(R \cup (\Delta_{0..4})\right)$$

Prove on **this side** that:

$$\left(M\left(RflxClsr(R)\right)\right)(0,1) = \left(M\left(R \cup (\Delta_{0..4})\right)\right)(0,1)$$

17 (1abc) Carry out the following instructions

1 (vi) Draw:

$$DG(R), \qquad DG(SymClsr(R)) \text{ and}$$

find: $$M\left(SymClsr(R)\right), M\left(R \cup (R^{op})\right)$$

Prove on **this side** that:

$$\left(M\left(SymClsr(R)\right)\right)(0,1) = \left(M\left(R \cup (R^{op})\right)\right)(0,1)$$

17 (1abc) Carry out the following instructions

1 (vii) Draw:

$$DG(R), \qquad DG(TrnsClsr(R)) \qquad \text{and}$$

find: $$M\left(TrnsClsr(R)\right), M\left(\bigcup_{k=1}^{5}(R^{\circ k})\right)$$

Prove on **this side** that:

$$\left(M\left(TrnsClsr(R)\right)\right)(0,1) = \left(M\left(\bigcup_{k=1}^{5}(R^{\circ k})\right)\right)(0,1)$$

17 (1abc) Carry out the following instructions

1 (viii) Draw:

$$DG(R), \qquad DG\left(EqvClsr(R)\right) \text{ and}$$

find: $EqvClsr(R), \qquad M\left(EqvClsr(R)\right)$

Record on **this side**:

$$\left\{EqvCls(x,R) \in \mathcal{P}(0..4) \middle| x \in \{0,1,2,3,4\}\right\}$$

$$= \quad \frac{\{0,1,2,3,4\}}{EqvClsr(R)}$$

$$=$$

17 (1abc) Carry out the following instructions

5 (ix) Draw:

$$DG(R), \qquad DG(EqvClsr(R)) \text{ and}$$

find: $$M\left(EqvClsr(R)\right), \; M\left(\bigcup_{k=0}^{5}\left(\left(R \cup (R^{op})\right)^{\circ k}\right)\right)$$

Prove on **this side** that:

$$\left(M\left(EqvClsr(R)\right)\right)(0,1) = \left(M\left(\bigcup_{k=0}^{5}\left(\left(R \cup (R^{op})\right)^{\circ k}\right)\right)\right)(0,1)$$

6 (2a) ***A*, *B* are as defined on page 202.** Circle the correct choices and record your proofs and witnesses on **pages that follow, properly numbered**. Express your answers as arrow-diagrams. Display the sources and targets as sets with elements.

$$\boldsymbol{A} = \left\{ \quad \right\} \quad \boldsymbol{\nu}(\boldsymbol{A}) = \qquad \boldsymbol{B} = \left\{ \quad \right\} \quad \boldsymbol{\nu}(\boldsymbol{B}) =$$

1(i) $\exists \alpha \in Fnc(A,B)\left(\left(Inj(\alpha)\right) \wedge \left(Srj(\alpha)\right)\right)$ Y N Pf W

1(ii) $\exists \beta \in Fnc(A,B)\left(\left(\cancel{Inj}(\beta)\right) \wedge \left(Srj(\beta)\right)\right)$ Y N Pf W

1(iii) $\exists \gamma \in Fnc(A,B)\left(\left(Inj(\gamma)\right) \wedge \left(\cancel{Srj}(\gamma)\right)\right)$ Y N Pf W

1(iv) $\exists \delta \in Fnc(A,B)\left(\left(\cancel{Inj}(\delta)\right) \wedge \left(\cancel{Srj}(\delta)\right)\right)$ Y N Pf W

1 (v) Find: $Im(\delta) =$

1 (vi) **On the following page**, draw arrow-diagrams for **every adjusted, maximal, bijective restriction** of $\delta: A \longrightarrow B$ and count the total number of **such restrictions**: _____

6 (2b) Define the relations $R \subseteq J \times J$ and $S \subseteq E \times E$ as follows:

$$R \coloneqq \left\{ (j\,,k) \in J \times J \;\middle|\; 2|(|j-k|) \right\} \quad \subseteq \quad J \times J$$

$$S \coloneqq \left\{ (e\,,f) \in E \times E \;\middle|\; \left(e \geq f\right) \vee \left(e|f\right) \right\} \quad \subseteq \quad E \times E$$

and answer the following questions.

3 (i) Decide if $Eqv(R)$ showing work on the following side:

$Y \quad N \quad Pf \quad W$

If yes, then draw $DG(R)$ on this side.

$DG(R)$:

3 (ii) Decide if $PO(S)$ showing work on the following side:

$Y \quad N \quad Pf \quad W$

If yes, then draw $HsDgrm(S)$ on this side.

$HsDgrm(S)$:

5 (2c) **Circle the correct choice(s)** on the right and **prove your assertion** accordingly:

Given that $PO(T,\leq)$ $\forall\varphi\in Fnc(S,T)$ $PO\big(R(\varphi)\big)$

Y N Pf W

where the relation

$$R(\varphi)\coloneqq\left\{(u\,,v)\in S\times S\middle|\varphi(u)=\varphi(v)\right\}\subseteq S\times S$$

6 (3a) Define the relations $P \subseteq J \times J$ and $Q \subseteq E \times E$ as follows:

$$P \coloneqq \left\{ (j\,,k) \in J \times J \;\middle|\; j|k \right\} \quad \subseteq \quad J \times J$$

$$Q \coloneqq \left\{ (e\,,f) \in E \times E \;\middle|\; f \mid e \right\} \quad \subseteq \quad E \times E$$

and answer the following questions.

3 (i) Decide if $PO(P)$ showing work on the following side:

$Y \quad N \quad Pf \quad W$

If yes, then draw $HsDgrm(P)$ on this side.

$HsDgrm(P)$:

3 (ii) Decide if $PO(Q)$ showing work on the following side:

$Y \quad N \quad Pf \quad W$

If yes, then draw $HsDgrm(Q)$ on this side.

$HsDgrm(Q)$:

6 (3b) **Circle the correct choices** in the below and **provide an answer** accordingly **on pages that follow, properly numbered**. Given a set S, for every pair of equivalence relations $E, F \in EqvRlns\left(S, S\right)$:

3 (i) $Eqv\left(E \backslash F\right)$ Y N Pf W

3 (ii) $Eqv\left((E \backslash F) \cup (F \backslash E)\right)$ Y N Pf W

5 (3c) On **pages that follow, properly numbered**, draw digraphs, Hasse diagrams, or sets as appropriate for the following:

$PrtOrdrs(B)$ ≔ the set of partial orders on B

$TtlOrdrs(B)$ ≔ the set of total orders on B

$EqvRlns(B)$ ≔ the set of equivalence relations on B

$Prtns(B)$ ≔ the set of partitions on B

and, determine the following:

2 (i) $\nu\left(PrtOrdrs(B)\right) =$

$PrtOrdrs(B) =$

1 (ii) $\nu\left(TtlOrdrs(B)\right) =$

$TtlOrdrs(B) =$

1 (iii) $\nu\left(EqnRlns(B)\right) =$

$EqnRlns(B) =$

1 (iv) $\nu\left(Prtns(B)\right) =$

$Prtns(B) =$